The Bogoliubov structure factor under the dual-scale envelope $\sqrt{k/2k_s}$: kernel-checked shape theorems, a one-sided diffraction test, and a Gross–Pitaevskii linear-response check in rusty-SUNDIALS

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Abstract

For the zero-temperature weakly interacting Bose gas, the Bogoliubov static structure factor is $S_{\mathrm{B}}(k) = \varepsilon_0(k)/\varepsilon_{\mathrm{B}}(k)$, where $\varepsilon_0 = \hbar^2k^2/2m$ and $\varepsilon_{\mathrm{B}} = \sqrt{\varepsilon_0(\varepsilon_0 + 2gn)}$. With $k_s = 2mc/\hbar$ it equals $k/\sqrt{k_s^2 + k^2}$.

Tier A (kernel-checked, Lean v4.34.0-rc2 in the QuantumFluids project environment; axioms within {`propext`, `Classical.choice`, `Quot.sound`}; planted `sorry` and planted-axiom controls caught). We prove 26 statements about these model formulas:

- the envelope $S_{\mathrm{B}} \le \sqrt{k/2k_s}$, with equality exactly at $k = k_s$;
- strict monotonicity of S_{B};
- the two-sided phonon bound $k/(k_s + k) < S_{\mathrm{B}} < k/k_s$;
- $S_{\mathrm{B}} < 1$, the rate $1 - S_{\mathrm{B}} \le k_s^2/k^2$ and the limit $S_{\mathrm{B}} \to 1$;
- a *diffraction test*: a structure factor that rises above the envelope is not S_{B}; and, *given* Feynman's inequality as a hypothesis, the measured dispersion is then not Bogoliubov.

The test's controls are kernel-checked too: it fires on the free value $S = 1$ and never fires on S_{B}.

Tier X (floating point; supports no claim). A new linear-response module in the rusty-SUNDIALS crate `qf-gpe2d` recovers $\varepsilon_{\mathrm{B}}(k)$ from Gross–Pitaevskii dynamics to 4.09×10^{-5} relative error. All 14 numeric gates pass, including $g = 0$ and splitting negative controls. Along the way we found that the pinned upstream reference data are reproduced by a reused-phase composition, not by the symmetric Strang step; the upstream source itself was not inspected. The crate's own `step`/`evolve` use the same composition. In linear response that composition converges only at first order in the amplitude; a new true Strang option converges at second order.

Real data (Tier L inputs, Tier X evaluation). Using the published ^{4}He dispersion table, the threshold at the roton is $\sqrt{Q_R/2k_s} = 0.5662$. Assuming Feynman's inequality, the dispersion table alone says the test is silent for $k \le 0.90\,\text{Å}^{-1}$ and cannot decide from $0.95\,\text{Å}^{-1}$ upward, including the roton.

Unmet deliverable: HuggingFace. The task asked for a HuggingFace dataset to illustrate the result. That was not delivered: a search of 30 HuggingFace queries found no structure-factor dataset, so no HuggingFace data illustrate this paper. The published ^{4}He table, which is not on HuggingFace, is the real-data illustration instead.

There is no novelty in mathematics or physics: the core inequality is $(k - k_s)^2 \ge 0$. What is offered is a kernel-checked statement in the owner project's vocabulary, a tested solver extension, and honest bookkeeping.

Tiers. [A]: a Lean statement checked by the kernel, axioms within the allowlist, planted controls caught. A Tier A tag attaches to the Lean statement about the *model formulas*, never to its physical gloss. [L]: a cited source, quoted verbatim in the ledger with line numbers of the alphaXiv text extraction. Here L covers verbatim *data values and quotes*. That is looser than the Elenchus L tier, which accepts theorem statements only: Elenchus `citation_check.py` refuses BS-L-0001 under its honest kind `data_value` (`CITE_KIND`). With the kind forced to `theorem`, used only to test whether the quotes are verbatim, it accepts BS-L-0001 and BS-L-0004; that probe is not a Tier L verdict. The BS-L-0004 source file is an excerpt written by the producing agent, not the full extraction. All three cited papers were checked against fresh alphaXiv reads by an independent agent. [C]: an argument or human proof that is not machine-checked. [X]: floating-point exploration, which never supports a claim. Identifiers such as `BS-A-0001` refer to `ledger/ledger.json` (Section 8).

Dependencies. Nothing here depends on the openai/math release or on any triangular-lattice theorem. The conditionalities are different, and they are stated where they enter:

1. the zero-temperature Bogoliubov model formulas, taken as *definitions* in Lean;
2. Feynman's inequality $\varepsilon(k) \le \varepsilon_0(k)/S(k)$, which enters only as the explicit hypothesis `hfeyn` of `diffraction_test_dispersion`;
3. the measured sound speed c, which sets k_s whenever the test is applied to data.

1 System, units and regime

Consider a uniform three-dimensional Bose gas of density n and mass m with contact coupling g, at $T = 0$, in the weak-coupling (Bogoliubov) regime $na^3 \ll 1$. The sound speed is $c = \sqrt{gn/m}$ and the healing length is $\xi = \hbar/(\sqrt{2}\,mc)$. We write

$$k_s := \frac{2mc}{\hbar} = \frac{\sqrt{2}}{\xi}, \qquad \hbar c =: h_c.$$

Then $\varepsilon_0(k) = \hbar^2 k^2/2m = h_c k^2/k_s$ and $2gn\,\varepsilon_0 = (h_c k)^2$, so the Bogoliubov dispersion is

$$\varepsilon_{\mathrm{B}}(k)^2 = \varepsilon_0(\varepsilon_0 + 2gn) = (h_c k)^2 + (h_c k^2/k_s)^2. \tag{1}$$

This is exactly the owner project's `bogoliubovSq hc ks k` (Listing 1) [C, BS-C-0002]. The Bogoliubov static structure factor is $S_{\mathrm{B}} = (u_k - v_k)^2 = \varepsilon_0/\varepsilon_{\mathrm{B}}$ [C, BS-C-0002]. In Lean it is a *definition*, so the Bijl–Feynman relation $\varepsilon_{\mathrm{B}} = \varepsilon_0/S_{\mathrm{B}}$ holds by construction.

The quantity $\sqrt{k/2k_s}$ comes from the owner's "dual-scale" reading of the dispersion. Write $\ell(k) = \varepsilon(k)^2/(h_c^2 k^3)$. Bogoliubov gives $\ell = 1/k + k/k_s^2 \ge 2/k_s$ (`ell_ge`). For a Feynman spectrum ε_0/S the same bound is equivalent to $S^2 \le k/2k_s$ (`feynman_ge_iff`). The owner project proves both. It does not name S_{B}, and it does not state $S_{\mathrm{B}} \le \sqrt{k/2k_s}$.

What the theorems say and do not say. They are statements about the formulas above. They say nothing about any real fluid unless the formulas describe it. Applied to data they become a *one-sided* test: a measured $S(k)$ above $\sqrt{k/2k_s}$ at some k is incompatible with the Bogoliubov S_{B} of the same k_s. A measured S below the envelope proves nothing.

2 Step-by-step human demonstration

This section is written for a strong undergraduate. Every step is proved by hand and then paired with the Lean declaration that checks it [A, BS-A-0001 to BS-A-0008]. Throughout, $h_c > 0$, $k > 0$ and $k_s > 0$, and $x := k/k_s$.

Step 1: the dispersion in project variables. By (1), $\varepsilon_{\mathrm{B}}^2 = \varepsilon_0^2 + (h_c k)^2 > 0$ because $h_c k > 0$. Lean: `bogoliubovSq_pos`, `epsBog_sq`.

Step 2: closed form. $S_{\mathrm{B}}^2 = \varepsilon_0^2/\varepsilon_{\mathrm{B}}^2$. Divide numerator and denominator by $h_c^2 k^2/k_s^2$:

$$S_{\mathrm{B}}^2 = \frac{h_c^2 k^4/k_s^2}{h_c^2 k^2 + h_c^2 k^4/k_s^2} = \frac{k^2}{k_s^2 + k^2}.$$

h_c drops out. Since $S_{\mathrm{B}} > 0$ (a quotient of positive numbers), $S_{\mathrm{B}} = k/\sqrt{k_s^2 + k^2}$. Lean: `sBog_pos`, `sBog_sq`, `sBog_eq_closed`.

Step 3: the envelope. Both sides are positive, so $S_{\mathrm{B}}^2 \le k/(2k_s) \iff 2k_s k^2 \le k(k_s^2 + k^2) \iff 0 \le k(k - k_s)^2$. The last inequality is true. Taking square roots, which preserve order on $[0, \infty)$, gives $S_{\mathrm{B}} \le \sqrt{k/2k_s}$. Lean: `sBog_sq_le`, `sBog_le_sqrt`.

Step 4: sharpness. Equality in Step 3 holds iff $k(k - k_s)^2 = 0$. Since $k > 0$, that means $k = k_s$, the self-dual wavenumber. Lean: `sBog_sq_eq_iff`. The hypotheses can be met, and the equality case occurs: $S_{\mathrm{B}}(1)^2 = 1/2$ for $h_c = k_s = 1$ (`nonvacuous_selfdual`).

Step 5: the second route through the owner's lemmas. The Feynman energy built from S_{B} is $\varepsilon_0/S_{\mathrm{B}} = \varepsilon_{\mathrm{B}}$ (by construction, `feynman_sBog_exact`). So $\ell = 1/k + k/k_s^2 \ge 2/k_s$, since $1/k + k/k_s^2 - 2/k_s = (k - k_s)^2/(k k_s^2)$. By `feynman_ge_iff` this is again $S_{\mathrm{B}}^2 \le k/2k_s$. Lean: `sBog_sq_le_via_project`. This route checks that the new statement agrees with the project's existing vocabulary. It is not independent evidence.

Step 6: monotonicity. $S_{\mathrm{B}}^2 = 1 - k_s^2/(k_s^2 + k^2)$ is strictly increasing in k. Positive numbers with increasing squares are increasing. Lean: `sBog_sq_lt_sq`, `sBog_strictMonoOn`.

Step 7: phonon regime. $k_s < \sqrt{k_s^2 + k^2} < k_s + k$, where the right inequality is $k_s^2 + k^2 < k_s^2 + 2k_s k + k^2$. Dividing k by these gives

$$\frac{k}{k_s + k} < S_{\mathrm{B}} < \frac{k}{k_s} = \frac{\hbar k}{2mc}.$$

So $S_{\mathrm{B}} \sim \hbar k/2mc$ as $k \to 0$, with relative error below k/k_s. Lean: `phonon_lower_lt_sBog`, `sBog_lt_phonon`.

Step 8: approach to the free particle. $S_{\mathrm{B}} < 1$ because $k < \sqrt{k_s^2 + k^2}$. Since $0 < S_{\mathrm{B}} < 1$, $1 - S_{\mathrm{B}} \le (1 - S_{\mathrm{B}})(1 + S_{\mathrm{B}}) = 1 - S_{\mathrm{B}}^2 = k_s^2/(k_s^2 + k^2) \le k_s^2/k^2$. The bound tends to 0, so $S_{\mathrm{B}} \to 1$ as $k \to \infty$. Lean: `sBog_lt_one`, `one_sub_sBog_le`, `sBog_tendsto_one`, and `epsBog_eq_epsFree_div_sBog` for $\varepsilon_{\mathrm{B}} = \varepsilon_0/S_{\mathrm{B}}$.

Step 9: the diffraction test, model level. Step 3 contraposed: if $S > \sqrt{k/2k_s}$ then $S \ne S_{\mathrm{B}}(k)$. Lean: `diffraction_test_model`. For $S > 0$, $\sqrt{k/2k_s} < S \iff k/2k_s < S^2$ (`peak_sqrt_iff_sq`).

Step 10: Feynman's inequality (not formalized). [C, BS-C-0001.] At $T = 0$, let $S(k, \omega) \ge 0$ be the dynamic structure factor per particle. Its zeroth moment is $\int S(k, \omega)\, d\omega = S(k)$, and the f-sum rule gives $\int \hbar\omega\, S(k, \omega)\, d\omega = \varepsilon_0(k)$. If $\varepsilon(k)$ is the lowest energy of a state reached from the ground state by the density operator ρ_k, then $S(k, \omega)$ vanishes for $\hbar\omega < \varepsilon(k)$, so

$$\varepsilon(k)\, S(k) \le \int \hbar\omega\, S(k, \omega)\, d\omega = \varepsilon_0(k), \qquad \text{i.e.} \qquad \varepsilon(k) \le \varepsilon_0(k)/S(k).$$

This is the inequality usually attributed to Feynman [4]. It is used here only as a stated hypothesis. Identifying the measured single-excitation branch with the lowest density-channel energy is a further assumption (Section 5).

Step 11: the diffraction test, dispersion level. Assume $\varepsilon^2 \le (\varepsilon_0/S)^2$ (the hypothesis `hfeyn`) and $S^2 > k/2k_s$. Then

$$\varepsilon^2 \le \frac{\varepsilon_0^2}{S^2} < \varepsilon_0^2 \cdot \frac{2k_s}{k} = \frac{2h_c^2 k^3}{k_s} \le h_c^2 k^2 \Big(1 + \frac{k^2}{k_s^2}\Big) = \varepsilon_{\mathrm{B}}^2,$$

where the last step is $2x \le 1 + x^2$. So $\varepsilon^2 \ne \varepsilon_{\mathrm{B}}^2$: the dispersion is not Bogoliubov. Lean: `diffraction_test_dispersion`, which reduces to the owner's `not_bogoliubov_of_structure_factor`.

Step 12: controls of the test. *Positive:* the free value $S = 1$ makes the test fire at $k = k_s$, since $\sqrt{1/2} < 1$. More generally $\sqrt{k/2k_s} < 1$ for $k < 2k_s$, and $k/k_s < 1$ for $k < k_s$, which breaks the phonon bound. Lean: `diffraction_fires_free_at_selfdual`, `free_S_one_violates`, `free_S_one_violates_phonon`. *Negative:* the test never fires on S_{B} itself, by Step 3. Lean: `diffraction_silent_on_sBog`. A caveat from the round-1 verifier: $S = 1$ is the $c \to 0$ limit, where $k_s \to 0$ too. The control fixes k_s, so it is a weak control.

3 The Lean formalization

Files and toolchain. The claimed module is `results/autoresearch/2026-10-25/H8/contribution/Autoresearch_SkBoundShape.lean` (sha256 fac6c85f9e6c…), in namespace `QuantumFluids.AutoresearchSkBoundShape`. Planted controls live only in `Autoresearch_SkBoundShapeControls.lean` (sha256 0beb41c99980…). The module imports the owner's compiled `DualLength` from SocrateAI-Scientific-QuantumFluids at commit `10f74daf1b39d2ac51a785452329fff0003cfc83` (`lean_src/DualLength.lean`, sha256 a28ce47e70c6…). The toolchain is Lean v4.34.0-rc2 with Mathlib `85e3a25e006c35636f0e53b0e9296caca2685bc0` (qf-lake), invoked directly as `lean` with `LEAN_PATH` set to the qf-lake build and package libraries. The file does not import Mathlib itself.

Gate record.

- Producer compile (Stage 1): rc 0 in 1004.9 s, no warnings or errors, 26 `#print axioms` lines, all within the allowlist. Earlier attempts timed out on disk I/O starvation, and one failed with two proof errors that were then fixed; all are kept in `contribution/attempt*.json`.
- Paper-stage recompile, by a separate agent instance using the producer's runner on the unchanged file (same sha256): main rc 0 in 12.1 s, 26 lines, nothing outside the allowlist; controls rc 0 in 9.1 s, both controls flagged. This is a kernel re-check by another AI agent instance. It is *not* a human audit and not a statement-adequacy review.
- The claimed file contains no `sorry`, `admit`, `native_decide` or `axiom` declaration (asserted by `gen_inputs.py`).
- The round-1 file `H8/Autoresearch_SkBound.lean` was recompiled by an independent verifier agent with verdict ACCEPT_WITH_CORRECTIONS. Its KEEP was obtained on a third, infrastructure retry outside the preregistered two-attempt rule, and is labelled that way.

The controls file exits rc 0 even though it contains a `sorry` and a planted axiom. That is why the axiom audit, not the exit code, is the gate.

Listing 1: Owner definitions and lemmas used (QuantumFluids `DualLength.lean`; proofs elided).

```
noncomputable def bogoliubovSq (hc ks k : ℝ) : ℝ := (hc * k) ^ 2 + (hc * k ^ 2 / ks) ^ 2

noncomputable def ell (ks k : ℝ) : ℝ := 1 / k + k / ks ^ 2

noncomputable def dualLength (hc k epsSq : ℝ) : ℝ := epsSq / (hc ^ 2 * k ^ 3)

theorem ell_ge {ks k : ℝ} (hk : 0 < k) (hks : 0 < ks) : 2 / ks ≤ ell ks k := -- proof elided

theorem feynman_ge_iff {ks k S : ℝ} (_hk : 0 < k) (hks : 0 < ks) (hS : 0 < S) :
    2 / ks ≤ k / (ks ^ 2 * S ^ 2) ↔ S ^ 2 ≤ k / (2 * ks) := -- proof elided

theorem not_bogoliubov_of_structure_factor {hc ks k S epsSq : ℝ}
    (hhc : hc ≠ 0) (hk : 0 < k) (hks : 0 < ks) (hS : 0 < S)
    (hfeyn : epsSq ≤ (hc * k ^ 2 / (ks * S)) ^ 2)        -- Feynman's variational bound
    (hpeak : k / (2 * ks) < S ^ 2) :                      -- measured: S above the dual-scale bound
    epsSq ≠ bogoliubovSq hc ks k := -- proof elided
```

Listing 2: Definitions in the claimed module (verbatim).

```
noncomputable def epsFree (hc ks k : ℝ) : ℝ := hc * k ^ 2 / ks

noncomputable def epsBog (hc ks k : ℝ) : ℝ := Real.sqrt (bogoliubovSq hc ks k)

noncomputable def sBog (hc ks k : ℝ) : ℝ := epsFree hc ks k / epsBog hc ks k
```

Listing 3: Round-1 theorems, re-proved under the same names (statements verbatim, proofs elided).

```
theorem bogoliubovSq_pos {hc ks k : ℝ} (hhc : 0 < hc) (hk : 0 < k) :
    0 < bogoliubovSq hc ks k := -- proof elided

theorem epsBog_sq {hc ks k : ℝ} (hhc : 0 < hc) (hk : 0 < k) :
    epsBog hc ks k ^ 2 = epsFree hc ks k ^ 2 + (hc * k) ^ 2 := -- proof elided

theorem sBog_pos {hc ks k : ℝ} (hhc : 0 < hc) (hk : 0 < k) (hks : 0 < ks) :
    0 < sBog hc ks k := -- proof elided

theorem sBog_sq {hc ks k : ℝ} (hhc : 0 < hc) (hk : 0 < k) (hks : 0 < ks) :
    sBog hc ks k ^ 2 = k ^ 2 / (ks ^ 2 + k ^ 2) := -- proof elided

theorem sBog_sq_le {hc ks k : ℝ} (hhc : 0 < hc) (hk : 0 < k) (hks : 0 < ks) :
    sBog hc ks k ^ 2 ≤ k / (2 * ks) := -- proof elided

theorem sBog_le_sqrt {hc ks k : ℝ} (hhc : 0 < hc) (hk : 0 < k) (hks : 0 < ks) :
    sBog hc ks k ≤ Real.sqrt (k / (2 * ks)) := -- proof elided

theorem sBog_sq_eq_iff {hc ks k : ℝ} (hhc : 0 < hc) (hk : 0 < k) (hks : 0 < ks) :
    sBog hc ks k ^ 2 = k / (2 * ks) ↔ k = ks := -- proof elided

theorem feynman_sBog_exact {hc ks k : ℝ} (hhc : 0 < hc) (hk : 0 < k) (hks : 0 < ks) :
    (hc * k ^ 2 / (ks * sBog hc ks k)) ^ 2 = bogoliubovSq hc ks k := -- proof elided

theorem sBog_sq_le_via_project {hc ks k : ℝ} (hhc : 0 < hc) (hk : 0 < k) (hks : 0 < ks) :
    sBog hc ks k ^ 2 ≤ k / (2 * ks) := -- proof elided

theorem free_S_one_violates {ks k : ℝ} (_hk : 0 < k) (hks : 0 < ks) (hlt : k < 2 * ks) :
    Real.sqrt (k / (2 * ks)) < 1 := -- proof elided
```

Listing 4: Shape theorems (statements verbatim, proofs elided).

```
theorem sBog_eq_closed {hc ks k : ℝ} (hhc : 0 < hc) (hk : 0 < k) (hks : 0 < ks) :
    sBog hc ks k = k / Real.sqrt (ks ^ 2 + k ^ 2) := -- proof elided

theorem sBog_sq_lt_sq {hc ks a b : ℝ} (hhc : 0 < hc) (hks : 0 < ks) (ha : 0 < a) (hab : a < b) :
    sBog hc ks a ^ 2 < sBog hc ks b ^ 2 := -- proof elided

theorem sBog_strictMonoOn {hc ks : ℝ} (hhc : 0 < hc) (hks : 0 < ks) :
    StrictMonoOn (sBog hc ks) (Set.Ioi 0) := -- proof elided
```

```
theorem sBog_lt_phonon {hc ks k : ℝ} (hhc : 0 < hc) (hk : 0 < k) (hks : 0 < ks) :
    sBog hc ks k < k / ks := -- proof elided

theorem phonon_lower_lt_sBog {hc ks k : ℝ} (hhc : 0 < hc) (hk : 0 < k) (hks : 0 < ks) :
    k / (ks + k) < sBog hc ks k := -- proof elided

theorem sBog_lt_one {hc ks k : ℝ} (hhc : 0 < hc) (hk : 0 < k) (hks : 0 < ks) :
    sBog hc ks k < 1 := -- proof elided

theorem one_sub_sBog_le {hc ks k : ℝ} (hhc : 0 < hc) (hk : 0 < k) (hks : 0 < ks) :
    1 - sBog hc ks k ≤ ks ^ 2 / k ^ 2 := -- proof elided

theorem sBog_tendsto_one {hc ks : ℝ} (hhc : 0 < hc) (hks : 0 < ks) :
    Filter.Tendsto (sBog hc ks) Filter.atTop (nhds 1) := -- proof elided

theorem epsBog_eq_epsFree_div_sBog {hc ks k : ℝ} (hhc : 0 < hc) (hk : 0 < k) (hks : 0 < ks) :
    epsBog hc ks k = epsFree hc ks k / sBog hc ks k := -- proof elided
```

Listing 5: The diffraction test and its in-file controls (statements verbatim, proofs elided).

```
theorem diffraction_test_model {hc ks k S : ℝ} (hhc : 0 < hc) (hk : 0 < k) (hks : 0 < ks)
    (hpeak : Real.sqrt (k / (2 * ks)) < S) : S ≠ sBog hc ks k := -- proof elided

theorem diffraction_test_dispersion {hc ks k S epsSq : ℝ} (hhc : 0 < hc) (hk : 0 < k)
    (hks : 0 < ks) (hS : 0 < S)
    (hfeyn : epsSq ≤ (hc * k ^ 2 / (ks * S)) ^ 2)
    (hpeak : Real.sqrt (k / (2 * ks)) < S) : epsSq ≠ bogoliubovSq hc ks k := -- proof elided

theorem peak_sqrt_iff_sq {ks k S : ℝ} (hS : 0 < S) :
    Real.sqrt (k / (2 * ks)) < S ↔ k / (2 * ks) < S ^ 2 := -- proof elided

theorem diffraction_fires_free_at_selfdual {hc ks : ℝ} (hhc : 0 < hc) (hks : 0 < ks) :
    (1 : ℝ) ≠ sBog hc ks ks := -- proof elided

theorem diffraction_silent_on_sBog {hc ks k : ℝ} (hhc : 0 < hc) (hk : 0 < k) (hks : 0 < ks) :
    ¬ (Real.sqrt (k / (2 * ks)) < sBog hc ks k) := -- proof elided

theorem free_S_one_violates_phonon {ks k : ℝ} (hks : 0 < ks) (hlt : k < ks) : k / ks < 1 := -- proof elided

theorem nonvacuous_selfdual : sBog 1 1 1 ^ 2 = 1 / (2 * 1) := -- proof elided
```

Listing 6: Planted controls file, in full. These are not claims.

```
/- Planted controls for the H8 shape file: the axiom audit must flag both.
Self-contained (re-declares `sBog` locally) so it does not need the claimed module's .olean. -/
import DualLength

namespace QuantumFluids.AutoresearchSkBoundShapeControls

open QuantumFluids.DualLength

noncomputable def sBog (hc ks k : ℝ) : ℝ :=
  (hc * k ^ 2 / ks) / Real.sqrt (bogoliubovSq hc ks k)

/-- Planted `sorry` on a FALSE shape claim (S_B ≥ 1): must show `sorryAx`. -/
theorem ctl_sorry {hc ks k : ℝ} (hhc : 0 < hc) (hk : 0 < k) (hks : 0 < ks) :
    1 ≤ sBog hc ks k := by
  sorry

/-- Planted axiom asserting the false claim that S_B is antitone: must show `ctl_cheat`. -/
axiom ctl_cheat : ∀ hc ks : ℝ, StrictAntiOn (sBog hc ks) (Set.Ioi 0)

theorem ctl_axiom {hc ks : ℝ} : StrictAntiOn (sBog hc ks) (Set.Ioi 0) := ctl_cheat hc ks

end QuantumFluids.AutoresearchSkBoundShapeControls

#print axioms QuantumFluids.AutoresearchSkBoundShapeControls.ctl_sorry
#print axioms QuantumFluids.AutoresearchSkBoundShapeControls.ctl_axiom
```

Axiom audit, paper-stage recompile (verbatim; Lean's line-wrap continuations joined).

```
'QuantumFluids.AutoresearchSkBoundShape.bogoliubovSq_pos' depends on axioms: [propext, Classical.choice, Quot.
    sound]
'QuantumFluids.AutoresearchSkBoundShape.epsBog_sq' depends on axioms: [propext, Classical.choice, Quot.sound]
'QuantumFluids.AutoresearchSkBoundShape.sBog_pos' depends on axioms: [propext, Classical.choice, Quot.sound]
'QuantumFluids.AutoresearchSkBoundShape.sBog_sq' depends on axioms: [propext, Classical.choice, Quot.sound]
'QuantumFluids.AutoresearchSkBoundShape.sBog_sq_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'QuantumFluids.AutoresearchSkBoundShape.sBog_le_sqrt' depends on axioms: [propext, Classical.choice, Quot.sound]
'QuantumFluids.AutoresearchSkBoundShape.sBog_sq_eq_iff' depends on axioms: [propext, Classical.choice, Quot.sound
    ]
'QuantumFluids.AutoresearchSkBoundShape.feynman_sBog_exact' depends on axioms: [propext, Classical.choice, Quot.
    sound]
'QuantumFluids.AutoresearchSkBoundShape.sBog_sq_le_via_project' depends on axioms: [propext, Classical.choice,
    Quot.sound]
'QuantumFluids.AutoresearchSkBoundShape.free_S_one_violates' depends on axioms: [propext, Classical.choice, Quot.
    sound]
'QuantumFluids.AutoresearchSkBoundShape.sBog_eq_closed' depends on axioms: [propext, Classical.choice, Quot.sound
    ]
'QuantumFluids.AutoresearchSkBoundShape.sBog_sq_lt_sq' depends on axioms: [propext, Classical.choice, Quot.sound]
'QuantumFluids.AutoresearchSkBoundShape.sBog_strictMonoOn' depends on axioms: [propext, Classical.choice, Quot.
    sound]
'QuantumFluids.AutoresearchSkBoundShape.sBog_lt_phonon' depends on axioms: [propext, Classical.choice, Quot.sound
    ]
'QuantumFluids.AutoresearchSkBoundShape.phonon_lower_lt_sBog' depends on axioms: [propext, Classical.choice, Quot.
    sound]
'QuantumFluids.AutoresearchSkBoundShape.sBog_lt_one' depends on axioms: [propext, Classical.choice, Quot.sound]
'QuantumFluids.AutoresearchSkBoundShape.one_sub_sBog_le' depends on axioms: [propext, Classical.choice, Quot.
    sound]
'QuantumFluids.AutoresearchSkBoundShape.sBog_tendsto_one' depends on axioms: [propext, Classical.choice, Quot.
    sound]
'QuantumFluids.AutoresearchSkBoundShape.epsBog_eq_epsFree_div_sBog' depends on axioms: [propext, Classical.choice,
     Quot.sound]
'QuantumFluids.AutoresearchSkBoundShape.diffraction_test_model' depends on axioms: [propext, Classical.choice,
    Quot.sound]
'QuantumFluids.AutoresearchSkBoundShape.diffraction_test_dispersion' depends on axioms: [propext, Classical.
    choice, Quot.sound]
'QuantumFluids.AutoresearchSkBoundShape.peak_sqrt_iff_sq' depends on axioms: [propext, Classical.choice, Quot.
    sound]
'QuantumFluids.AutoresearchSkBoundShape.diffraction_fires_free_at_selfdual' depends on axioms: [propext,
    Classical.choice, Quot.sound]
'QuantumFluids.AutoresearchSkBoundShape.diffraction_silent_on_sBog' depends on axioms: [propext, Classical.choice,
     Quot.sound]
'QuantumFluids.AutoresearchSkBoundShape.free_S_one_violates_phonon' depends on axioms: [propext, Classical.choice,
     Quot.sound]
'QuantumFluids.AutoresearchSkBoundShape.nonvacuous_selfdual' depends on axioms: [propext, Classical.choice, Quot.
    sound]
```

Planted controls, paper-stage recompile (same convention).

```
'QuantumFluids.AutoresearchSkBoundShapeControls.ctl_sorry' depends on axioms: [propext, sorryAx, Classical.choice,
     Quot.sound]
'QuantumFluids.AutoresearchSkBoundShapeControls.ctl_axiom' depends on axioms: [propext, Classical.choice, Quot.
    sound, QuantumFluids.AutoresearchSkBoundShapeControls.ctl_cheat]
```

`gen_inputs.py` extracts every listing from the source files, byte for byte, into `gen/*.lean`. The typeset `gen/*.lst` copies differ only in that non-ASCII symbols are escaped to LaTeX glyphs.

4 Numerics in rusty-SUNDIALS

All numbers in this section are [X]: floating-point, producer-checked only, supporting no claim.

Method. The crate `qf-gpe2d` integrates the 2D Gross–Pitaevskii equation $i\psi_t = -\frac{1}{2}\nabla^2\psi + g|\psi|^2\psi - \mu\psi$ by Fourier split-step on a periodic grid, with $\hbar = m = 1$, $n_0 = g = 1$, so $c = 1$, $k_s = 2$ and $\xi = 1/\sqrt{2}$. The new module `linear_response`:

1. relaxes the uniform ground state in imaginary time from a perturbed start (max $||\psi|^2 - n_0| = 4.31 \times 10^{-14}$);

2. applies a weak phase kick $\psi \to \psi\, e^{i\epsilon \cos k\cdot r}$ or density kick $\psi \to \psi\sqrt{1+\epsilon\cos k\cdot r}$ on one grid wavevector;

3. records the density cosine component $A(t)$ and reads ω from zero-crossing times and from a least-squares sinusoid fit.

Linearizing the GP equation gives $\omega^2 = \varepsilon_0(\varepsilon_0 + 2gn)$. After a phase kick, continuity gives $A(t) = (nk^2\epsilon/\omega)\sin\omega t$, so the amplitude yields a second estimate $S_{\rm amp} = A_{\max}/(2n\epsilon) = \varepsilon_0/\omega$ [C, BS-C-0003]. A classical GP field has no quantum density fluctuations, so $S = \varepsilon_0/\omega$ is the Bijl–Feynman identification, not an independent measurement of a quantum $S(k)$. This is the same by-construction point as `feynman_sBog_exact`.

The sweep uses a 32×32 grid, 25 wavevectors with k/k_s from 0.05 to 5, 6 periods, 200 steps per period and $\epsilon = 1 \times 10^{-3}$.

Finding: the pinned reference data are not Strang. The first sweep failed its own gates: the frequency error was 1.23×10^{-2}, and $S_{\rm amp}$ was up to 25% above the frequency estimate. Replaying the pinned 4×4 Python reference data in `tests/python_cross_check.rs` shows the cause (`which_strang.py`). A composition that reuses the start-of-step nonlinear phase on both sides of the kinetic step matches the pinned data to 2.5×10^{-16}. The upstream Python source (`solver2d.py`) is not on this machine and was not inspected, so the statement is about the pinned data, not about the upstream code. The crate's own `step`/`evolve` implement the same reused-phase composition. A textbook Strang step, which recomputes $|\psi|^2$ after the kinetic step, deviates by 0.155 [X, BS-X-0003]. In linear response the reference composition still converges at second order in the frequency, but only at first order in the amplitude (errors 2.65×10^{-2} down to 1.63×10^{-3} over four halvings, observed order 1.013). A new `Splitting::Strang` option fixes both. The existing `step`/`evolve` are unchanged, so the Python cross-check still passes. Our reading ([C, BS-C-0004]): the reused-phase step is a non-symmetric composition, whose eigenfrequency error is $O(\Delta t^2)$ while its invariant ellipse is distorted at $O(\Delta t)$. One long-wavelength run ($k = 0.1$) with the reference step also showed late-time growth from round-off. That was a single run, and its mechanism was not analysed.

Table 1: The 14 numeric gates (`linear_response_summary.json`); 14/14 pass. [X]

Gate	Value	Threshold
Positive controls		
$g = 1$ phase kick, max $\lvert\omega - \varepsilon_{\rm B}\rvert/\varepsilon_{\rm B}$	4.09×10^{-5}	$< 10^{-4}$
$g = 1$ density kick (independent readout)	4.06×10^{-5}	$< 10^{-4}$
f-sum amplitude vs period, max $\lvert S_{\rm amp} - S_{\rm freq}\rvert/S_{\rm freq}$	1.64×10^{-4}	$< 5 \times 10^{-3}$
Strang Δt refinement, min observed order	2.000	> 1.8
Isotropy, modes (2,0), (1,1), (2,2)	1.78×10^{-7}	$< 10^{-4}$
Ground-state residual	4.31×10^{-14}	$< 10^{-10}$
Negative controls		
$g = 0$: max $\lvert\omega - k^2/2\rvert/(k^2/2)$	7.79×10^{-13}	$< 10^{-9}$
$g = 0$ vs $g = 1$ $\varepsilon_{\rm B}$ for $k < k_s$, min relative distance	0.3355	> 0.1
Reference splitting, amplitude mismatch (recorded failure)	0.2497	$> 5 \times 10^{-3}$
Reference splitting, amplitude order	1.013	< 1.5
Bound level (mirrors the Lean controls)		
$g = 1$: max $S_{\rm freq}/\sqrt{k/2k_s} - 1$	-1.381×10^{-3}	$< 10^{-4}$
$g = 1$: $\lvert S_{\rm freq}/{\rm bnd} - 1\rvert$ near $k{=}k_s$ (x=1.0772; measured; exact theory 1.3809×10^{-3})	1.381×10^{-3}	$< 2 \times 10^{-3}$
$S = 1$ fires at every sweep $k < 2k_s$	true	—
$S = 1$ silent at every sweep $k > 2k_s$ (one-sided test)	true	—

The two bound gates in Table 1 are the same sweep point ($x = 1.0772$, where the maximum occurs), so their values are equal in magnitude. Both are measured S_{freq}, which differs from the exact theory there by 6.9×10^{-8}, the size of the frequency error at that point; neither gate compares theory with itself. At exactly $k = k_s$, with the finest step and $\epsilon = 10^{-4}$, $S_{\mathrm{freq}}/\sqrt{k/2k_s} - 1 = -3.56 \times 10^{-9}$. At $k = k_s$ the Strang frequency error falls from 1.27×10^{-6} to 3.56×10^{-9} over four halvings. An independent 1D numpy implementation agrees with the Rust code in ω to between 1.9×10^{-14} and 1.6×10^{-12}, and in amplitude to 2.4×10^{-9} (the two codes fit the amplitude at slightly different ω) [X, BS-X-0004]. The crate's test suite passes 15/15, and `cargo fmt -check` and `cargo clippy -all-targets` exit 0 [X, BS-X-0007].

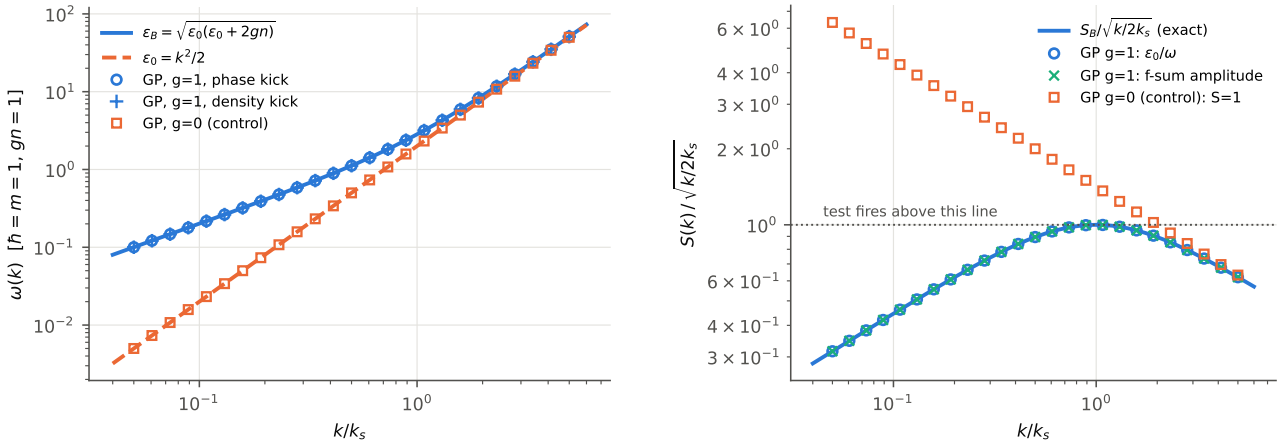


Figure 1: Left: $\omega(k)$ from GP linear response (phase and density kicks at $g = 1$, and the $g = 0$ control) against ε_{B} and ε_0. Right: $S/\sqrt{k/2k_s}$ for the exact S_{B}, the measured ε_0/ω, the f-sum amplitude and the $g = 0$ control, which exceeds 1 for $k < 2k_s$. Generated by `make_figures.py` from the result files. [X]

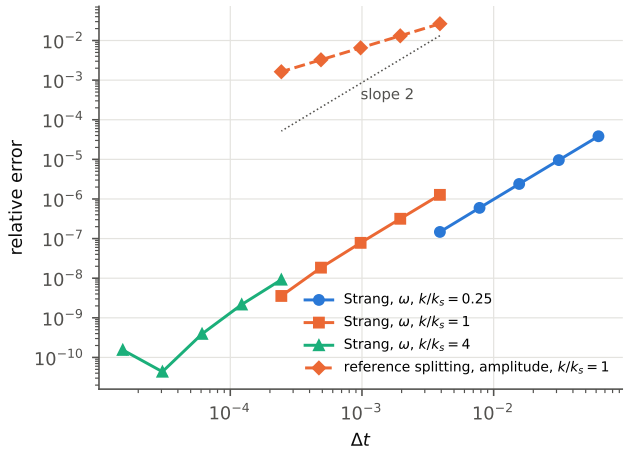


Figure 2: Strang second-order frequency convergence at three k, and the first-order amplitude error of the reference splitting. [X]

5 Real-data illustration: superfluid ^{4}He

Inputs [L]. The dispersion is Godfrin et al. [1], Table VI: $\varepsilon(k)$ at saturated vapour pressure and $T < 100\,\mathrm{mK}$, for $k = 0$ to $3.6\,\text{Å}^{-1}$ in steps of 0.05 (73 rows). Below $0.15\,\text{Å}^{-1}$ the entries are ultrasound data [L, BS-L-0002]. The sound speed is $c = 238.3 \pm 0.1\,\mathrm{m/s}$ and the molar mass is $4.0026032\,\mathrm{g/mol}$, from the same paper [L, BS-L-0001]. The roton cross-check is Beauvois et al. [2], Table I: $\Delta_R = 0.7416(10)\,\mathrm{meV}$ at $Q_R = 1.9260(2)\,\text{Å}^{-1}$ [L, BS-L-0003]. The table was transcribed from the alphaXiv text extraction. The transcription controls pass: uniform k spacing, a table minimum at (1.90, 0.7427) against the reference roton, and $\varepsilon(0.05)$ within 0.35% of $\hbar ck$.

Evaluation [X, BS-X-0005]. The scales are $k_s = 2m_4c/\hbar = 3.0038\,\text{Å}^{-1}$ and $\hbar^2/2m_4 = 0.52218\,\mathrm{meV\,Å^2}$. The threshold at the roton is $\sqrt{Q_R/2k_s} = 0.5662$, moving to 0.56609–0.56633 across the $\pm 0.1\,\mathrm{m/s}$ uncertainty in c.

A dispersion table cannot measure S. Under Feynman's inequality (Step 10) it gives only an upper bound $S(k) \le S_{\mathrm{up}}(k) = \varepsilon_0(k)/\varepsilon(k)$, computed conservatively from $\varepsilon - \mathrm{err}$ where Table VI gives an error bar. At $k = 0.05, 0.10\,\text{Å}^{-1}$ (ultrasound rows) it gives none, so there S_{up} uses ε itself and is not conservative; the margins are large (S_{up} vs threshold: 0.0166 vs 0.0912; 0.0329 vs 0.1290). For the 18 tabulated points $0.05 \le k \le 0.90\,\text{Å}^{-1}$, the computed S_{up} lies below $\sqrt{k/2k_s}$. So, *if* Feynman's inequality holds with the measured branch as the lowest density-channel energy, the test cannot fire on ^{4}He there. A stricter variant takes $\min(\varepsilon - \mathrm{err}, \hbar ck)$, allowing for multi-phonon states near $\hbar ck$ where the dispersion is anomalous. It leaves this set unchanged; at $k = 0.5\,\text{Å}^{-1}$, $S_{\mathrm{up}} = 0.1665$ against a threshold of 0.2885.

For the 54 points from $0.95\,\text{Å}^{-1}$ upward the table cannot decide. At the tabulated point nearest the roton, $k = 1.95\,\text{Å}^{-1}$, $S_{\mathrm{up}} = 2.66$ is far above the threshold there, $\sqrt{k/2k_s} = 0.5697$ (the threshold at $Q_R = 1.9260\,\text{Å}^{-1}$ itself is 0.5662), and only a measured, absolutely normalized $S(k)$ could decide. Because S_{up} is only an upper bound, a dispersion table can never make the test fire. **We therefore do not claim that the ^{4}He roton peak exceeds 0.5662.** We found no verified, absolutely normalized $S(Q)$ table. The cold-atom literature does not help for an absolute test: Shammass et al. [3] multiply their measured $S(k)$ by 1.5 "in order to agree with the 2D GPE simulation" [L, BS-L-0004]. Their short-Bragg-pulse standing-wave method is the experimental analogue of our phase kick.

Dispersion-level comparison [X, BS-X-0006]. This needs no S. With the same c, the measured $\varepsilon/\varepsilon_{\mathrm{B}}$ is 0.205 at $k = 1.95\,\text{Å}^{-1}$, with a minimum of 0.176 at $3.6\,\text{Å}^{-1}$. This is the standard fact that ^{4}He is not a weakly interacting gas, read directly from the pinned table. On the same code path, the Bogoliubov S_{B} built with the ^{4}He c never fires, and $S = 1$ fires at every tabulated $k < 2k_s$.

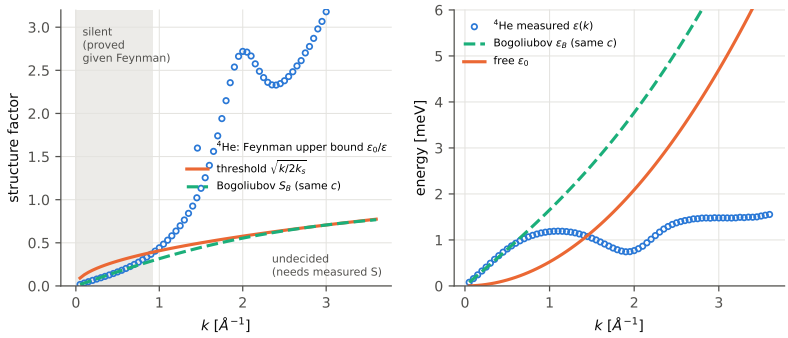


Figure 3: ^{4}He from Godfrin et al. Table VI. The Feynman upper bound $\varepsilon_0/\varepsilon$ is plotted against the threshold $\sqrt{k/2k_s}$, with the silent region shaded, alongside the Bogoliubov S_{B}, and measured ε against ε_{B} and ε_0. Data [L], evaluation [X].

6 HuggingFace dataset search: a null result

The task asked for a HuggingFace dataset to illustrate the result. We report the search as a null result rather than substitute an unrelated dataset [X, BS-X-0008].

- **Search.** 30 queries were run through `HfApi.list_datasets` (public metadata, no token). They included structure factor (static and dynamic), liquid and superfluid helium, neutron and X-ray diffraction, Bose–Einstein condensate, Bragg spectroscopy, cold atoms, Bogoliubov, pair and radial distribution function, phonon and SANS. The full record is `data/hf_search.json`.
- **Hits.** None of the hits is a measured $S(k)$, $S(Q,\omega)$, Bragg-spectroscopy or cold-atom dataset. The "helium" and "BEC" hits are name collisions (language-model checkpoints and other unrelated repositories).
- **The single plausible hit** is `popoff4rtem/2D-Neutron-Diffraction-Dataset-for-Discriminative-Models` at revision `648119c2030c5352c37a305deb74aa8ec84a4402`, tagged MIT. Its card holds only the licence header. Its data are Python pickles, the largest 4.38 GB, named for discriminative ML on 2D diffraction maps, and nothing identifies them as a liquid or ^{4}He $S(Q)$. Only the card and the file list were fetched (stored under `/home/callensxavier_gmail_com/hf-data/bogoliubov_sk/`). Nothing was downloaded or unpickled, and nothing was substituted.

The real-data illustration is therefore the published ^{4}He table of Section 5, which is not hosted on HuggingFace.

7 Limitations

- **Novelty.** There is none in mathematics or physics. The envelope is an AM–GM corollary of the owner's `feynman_ge_iff` and `ell_ge`.
- **By construction.** `feynman_sBog_exact` and `epsBog_eq_epsFree_div_sBog` hold because S_B is *defined* as $\varepsilon_0/\varepsilon_\mathrm{B}$. The GP $S = \varepsilon_0/\omega$ inherits this. Nothing here derives S_B from a many-body calculation.
- **Scope of the model.** The model is $T = 0$ Bogoliubov only, with no Beliaev damping and no LHY corrections.
- **Conditional application.** The dispersion-level test assumes Feynman's inequality ([C]), and applying it needs k_s from a measured c.
- **^{4}He.** The roton case is undecided. No absolutely normalized $S(Q)$ source was verified; Svensson et al. and similar diffraction work were not checked and are not cited. A cold-atom paper consulted at Stage 2 (arXiv cond-mat/0111438) could not be retrieved again at the paper stage and is not cited.
- **Verification.** Statement adequacy has been reviewed only by AI agents. The paper-stage recompile is a second kernel run by another agent instance, not a human audit. The ledger gate flags every Tier A row as unaudited.
- **Protocol and upstream.** The round-1 KEEP came from an out-of-protocol infrastructure retry. The reused-phase composition seen in the pinned reference data has not been reported upstream, and the upstream Python source was not inspected.
- **Numerics.** All numerics are 2D, classical-field and periodic. They are producer-checked only.

- **HuggingFace (unmet deliverable).** The requested HuggingFace illustration was not delivered: no Hub dataset holds a measured structure factor (Section 6). The ^{4}He table stands in as the real-data illustration.
- **Tier L scope.** Tier L here is verbatim data and quotes, looser than the Elenchus theorem-only L tier.

8 Claim ledger

Every tagged claim is a record in `ledger/ledger.json`, built by `ledger/build_ledger.py`. Each record points to a content-addressed evidence blob in `ledger/evidence/`:

- Tier A: verbatim axiom lines with source and transcript hashes;
- Tier L: verbatim excerpts with line numbers and the sha256 of the alphaXiv extraction, copied to `ledger/sources/`;
- Tier C: the argument text;
- Tier X: result files with `claim_licence: false`.

The Elenchus gate `tools/ledger.pyledger.json--evidence-direvidence` was run on the ledger. Its exit code and findings are recorded verbatim in `ledger/gate_output.txt`. The ledger holds 24 claims (8 A, 4 L, 4 C, 8 X). The gate exits 1 with 0 blocking findings and 8 flags, one `LEDGER_UNAUDITED_TIER_A` per Tier A row, because no person has audited statement adequacy. A first run (`ledger/gate_output_run1.txt`) also blocked a tier inversion: BS-C-0004 rested on the Tier X row BS-X-0003. We removed that dependency, so the reading stands as an unsupported argument. Bookkeeping that passes the gate licenses nothing about truth.

9 Reproducibility

- **Lean.** `python-Iresults/autoresearch/2026-10-25/H8/contribution/run_lean_rc2.pyFILE.leanOUT.json` from the AutoevolveAI worktree, for the claimed file and the controls file. The runner pins cores 4–7 and sets `LEAN_PATH` as above.
- **Rust.** rusty-SUNDIALS branch `physics/ar-bogoliubov-2026-10-25`, commits `e1449988a91d8a38a7deee7f1106cfa3962c5759` (code and results) and `fdd6d2d64fbab0dcaad60fa7b7877d96095c2776` (README, strict ^{4}He variant, test log) and `ae918316a82f7ac7102d53f646eb20c5a2c838a5` (HEAD; module-doc correction only, results unchanged), not pushed at the time of writing. `cargo build -release -p qf-gpe2d -examples`, then run `examples/bogoliubov_dispersion` (wall time 409 s, as recorded by the producing agent in `results/autoresearch/2026-10-25/H8/contribution/NUMERICS.md`; it is not written to a machine result file); scripts `summarize.py`, `crosscheck_compare.py`, `which_strang.py`, `helium_tier_l.py` and `make_figures.py` live in `crates/qf-gpe2d/results/bogoliubov_sk/`.
- **Paper.** `python -I gen_inputs.py` regenerates every number and listing from the result files. `python -I ledger/build_ledger.py` rebuilds the ledger. `python -I build_pdf.py` runs pdflatex twice.

References

[1] H. Godfrin, K. Beauvois, A. Sultan, E. Krotscheck, J. Dawidowski, B. Fåk, J. Ollivier, *The dispersion relation of Landau elementary excitations and the thermodynamic properties of superfluid ^{4}He*, arXiv:2012.09067v1 (2020).

[2] K. Beauvois, J. Dawidowski, B. Fåk, H. Godfrin, E. Krotscheck, J. Ollivier, A. Sultan, *Microscopic dynamics of superfluid ^{4}He: a comprehensive study by inelastic neutron scattering*, arXiv:1802.08120v3.

[3] I. Shammass, S. Rinott, A. Berkovitz, R. Schley, J. Steinhauer, *The phonon dispersion relation of a Bose–Einstein condensate*, arXiv:1207.3440.

[4] R. P. Feynman, Phys. Rev. **94**, 262 (1954). This is a bibliographic pointer only, taken from the reference lists of [1, 3]; the original was not read for this paper.

[5] SocrateAI-Scientific-QuantumFluids, `lean_src/DualLength.lean` at commit `10f74daf1b39d2ac51a785452329fff0003cfc83` (owner project; unpublished).

[6] X. Callens, *Planar universal optimality, tiered* (preprint v2), doi:10.5281/zenodo.23293429; and *The Triangular Lattice and Universal Optimality* (textbook), doi:10.5281/zenodo.23295142. These are earlier tiered publications of this programme, cited for method only.

The tier conventions and ledger format follow earlier tiered publications of this programme [6]. The owner lemmas come from [5].